Introduction to Einstein's General Relativity

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Abstract

These are Lecture Notes for PHYS 413/513 at Rice University. At the moment, they constitute a work in progress. Please report any typos/errors to mustafa.a.amin@rice.edu.

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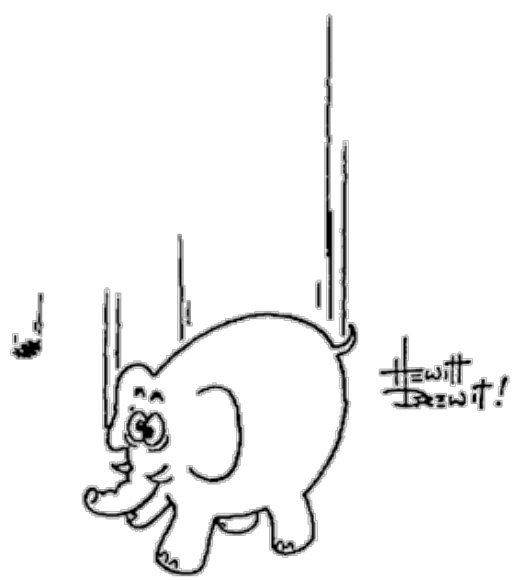

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Preface

These are notes for a one semester course, PHYS 413/513, at Rice University. The course is meant for students (undergraduate and graduate) who have little or no prior exposure to Einstein's General Relativity. However, the students are expected to have a solid foundation in classical mechanics (with Lagrangians), multivariable calculus, differential equations, and linear algebra. Familiarity with Special Relativity and Electrodynamics will be very helpful too.
Students: Do the recommended reading before class!

Acknowledgements

The primary textbook for the course is James Hartle's "Gravity". The notes rely partly on the textbook for organization, and are very much a work in progress. Don't expect complete notes at the moment. I have added my own examples, explanations and visualizations. where I thought I had something to say which can help students further. I will also supplement these notes with exercises to be done as you read through. This will all happen in due time.

I learnt General Relativity first by reading Sean Carroll's more mathematical textbook, which I thoroughly enjoyed. I was enamoured by the elegant differential geometry at the time, but missed some of the relevant physics in that first pass. Hartle's book helped address that for me. I have also benefited from Misner, Thorne and Wheeler's "Gravitation", Weinberg's presentation in "Gravitation and Cosmology" and Wald's book "General Relativity". More recently, I have enjoyed learning from the Special and General Relativity sections of Thorne and Blandford's "Modern Classical Physics".

I owe an immense gratitude to my teachers – Prof. Eva Silverstein who first asked me to learn GR during a "rotation" during the first year of graduate school, Prof. Renata Kallosh for my first course, and Prof. Robert Wagoner for letting me TA for his GR class. This is how I learnt GR during graduate school. Prof. Barbara Shipman at the University of Texas at Arlington taught me multivariable Calculus and Linear Algebra, and then differential Geometry during my undergraduate years, both in class and often on Friday afternoons beyond class. Her enthusiasm for and pedagogy was infectious, and I hope I can follow in her footsteps.

And, importantly, the previous Phys 413/513 students at Rice were critical in making the course what it is now.

In preparing these notes, I am using some LaTeX and editing assistance from ChatGPT.

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CHAPTER 1

INVITATION

We will explore one of the most impressive achievements of our species. Along the way, I will also tell you how to lose some weight.

1.1 Going along with Gravity

Launch a coconut and a tomato with the same velocity from the earth's surface. In absence of air resistance, they will follow the same trajectory![1] This, in itself is surprising – the trajectory is independent of their mass. If you were launched with them with the same velocity, they would simply appear to float next to you – gravity seems to have disappeared! More generally, if you were falling freely under gravity, by looking at the tomato and coconut (for a short enough period of time) they would each appear to be moving at a constant velocity, with no gravitational acceleration. Even if you, the coconut and tomato were launched with different individual initial velocities, you would still find that they appear to move at a constant velocity relative to you and each other. You will also "feel" weightless.

Exercise 1.1.1 Why is it surprising that in the gravitational field $\mathbf{g}$ the coconut and tomato, which have different masses m_{c} and m_{t}, follow the same trajectory? Here $|\mathbf{g}| = |-\nabla\Phi| \approx GM_{\mathrm{e}}/R_{\mathrm{e}}^2$. Consider the situation where the (approximately) constant gravitational field $\mathbf{g}$ near the earth surface is replaced by a constant electric field $\mathbf{E}$, and assume that the fruits have charges q_{c} and q_{t}. The accelerations would be $\mathbf{a}_{\mathrm{c}} \propto q_{\mathrm{c}}/m_{\mathrm{c}}$, and similarly $\mathbf{a}_{\mathrm{t}} \propto q_{\mathrm{t}}/m_{\mathrm{t}}$. It depends on the ratio q_i/m_i, ie. on the nature of the bodies, so the trajectories will be different! However, in the case of gravity, $\mathbf{a}_{\mathrm{c}} = \mathbf{a}_{\mathrm{t}} = \mathbf{g}$ independent of the mass! Check these statements using Newton's second law, as well as gravitational and electrostatic forces. See Fig. 1.1

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[1] Check out a version of this experiment on the moon `https://www.youtube.com/watch?v=KDp1tiUsZw8`. Also see `https://www.youtube.com/watch?v=E43-CfukEgs`.

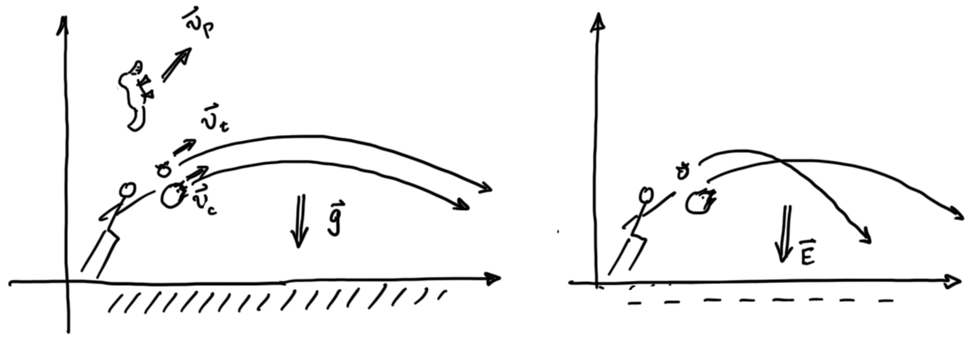

Figure 1.1

Exercise 1.1.2 Suppose that the fruit were launched from the surface of the earth with velocity $\mathbf{v}_\mathrm{c}$ and $\mathbf{v}_\mathrm{t}$. They will follow parabolic trajectories $\boldsymbol{x}_\mathrm{c}(t)$ and $\boldsymbol{x}_\mathrm{t}(t)$, with velocity $\mathbf{v}_\mathrm{c}(t)$ and $\mathbf{v}_\mathrm{t}(t)$ and acceleration $\mathbf{a}_\mathrm{c}(t) = \mathbf{a}_\mathrm{t}(t) = \mathbf{g}$. (i) Write down $\boldsymbol{x}_\mathrm{c,t}(t)$ and $\mathbf{v}_\mathrm{c,t}(t)$. Now imagine at the same time, a platypus named Polly starts falling freely from some height h, (ii) write down $\boldsymbol{x}_\mathrm{P}(t)$,$\mathbf{v}_\mathrm{P}(t)$ and $\mathbf{a}_\mathrm{P}(t)$. (iii) What would the position, velocity and acceleration of the coconut and tomato be as measured in the "falling frame" of Polly? You should find that $\mathbf{g}$ disappears; Polly will conclude there is no gravity in the situation. (iv) Let the initial velocities $\mathbf{v}_c = \mathbf{v}_P = 0$. The coconut and Polly are falling from a height $z_\mathrm{c}(t=0) = h_\mathrm{c}$ and $z_\mathrm{P}(t=0) = h_\mathrm{P}$, both starting at rest. In this case make a sketch of t vs. $z_c(t)$ in the earth's rest frame. Then make the same sketch in Polly's rest frame. What do you notice? Repeat the same exercise with non-zero, and different initial velocities for Polly and the coconut.

While not obvious to us on earth easily (unless you are the free, but unfortunate Polly), get in touch with your favorite "freely falling" astronaut in the international space station to understand how gravitational effects (mostly) disappear for them in their "local" laboratory on the space-station.[2]

[2] In 2024, I read a sign at the Johnson Space Center (inside the mock shuttle), which said astronauts float because of lack of earth's gravity since they are far away from the earth. After this course, you should tell me what you you think about this statement.

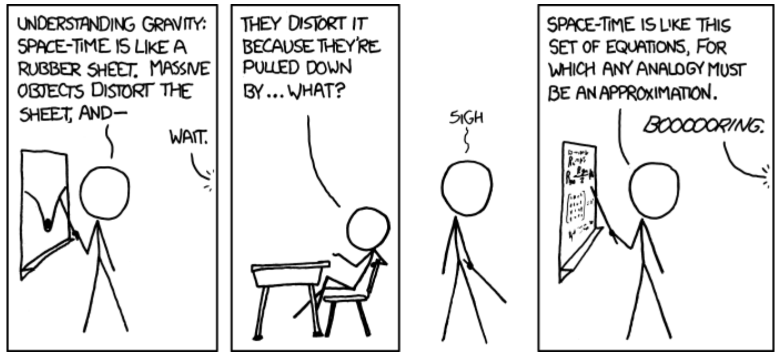

Figure 1.2: Credit: https://xkcd.com

With Einstein as our guide, these simple observations lead us to an astonishing insight into Nature. Gravity is unlike other known forces – it is really about the nature of space and time. Three dimensional space is not a passive arena in which events take place in time, but space and time together – *spacetime* – is an active, dynamical entity. It is a smooth collection of events: a 3+1 dimensional spacetime, whose geometry is determined by the mass-energy within it. The mass of the earth "curves" the spacetime around it.

As we will learn, despite the parabolic locus of the trajectory in space, the coconut and tomato are moving in this curved *spacetime* on the "straightest" possible path which is independent of the composition and mass of the fruit. Gravity is a manifestation of this curved spacetime.

1.2 The road ahead

We will spend much of the course developing a quantitative, calculable and testable understanding of the above ideas. Our underlying theoretical framework for gravity is that of Einstein's General Theory of Relativity (GR). This theory, arguably one of humanity's most elegant and well tested ideas, is built upon Einstein's Special Theory of Relativity. We will review Einstein's Special Theory of Relativity (SR) first – with an emphasis on geometry. The path to GR then is a matter of living with SR "locally" in small enough regions of spacetime, and developing an understanding of how to connect the happenings in many such small regions together. We will compare and contrast our developing understanding of GR with Newtonian ideas of gravity, and see why Newtonian ideas fail to be consistent with SR and more importantly, with observational and experimental tests. Nevertheless, it will be helpful to keep the following connection between Newtonian and Einstein gravity at the back of our mind: $\ddot{\boldsymbol{x}} = -\nabla\Phi \rightarrow$ geodesic equation, $\nabla^2\Phi = 4\pi G\rho \rightarrow$ Einstein Equations.

1.3 Newton vs. Einstein

The Newtonian theory of gravity is of course still extremely useful and accurate for most applications on the earth, solar system and in astrophysics more generally. However, Newtonian gravity fails when considering systems with mass M and extent R, such that GM/Rc^2 becomes order unity. Here $c = 3\times10^{10}\mathrm{cm\,s^{-1}}$ is the speed of light in vacuum and $G = 6.67\times10^{-8}\,\mathrm{dyne\,gm^{-2}\,cm^2}$

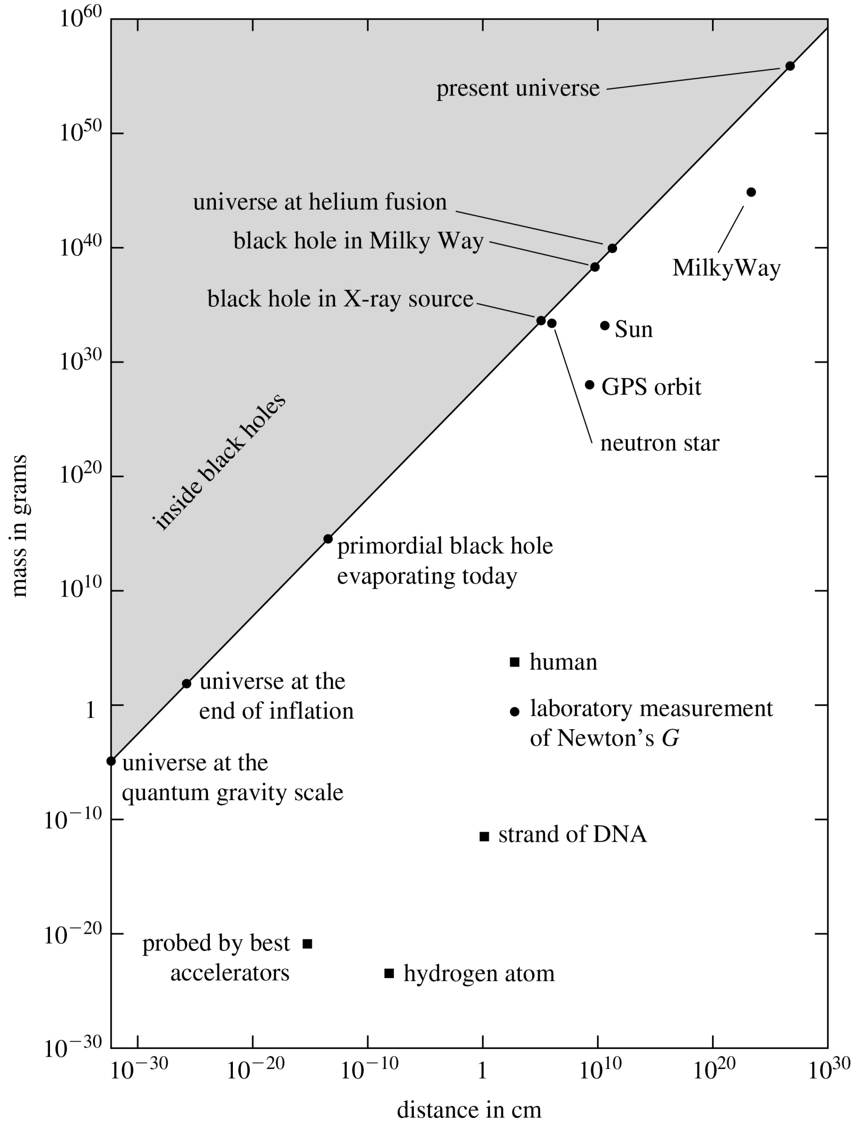

Figure 1.3: General relativity effects are not ignorable near the diagonal line, where $2GM/Rc^2 = 1$. Figure from J. Hartle. To understand the diagonal line, recall that the escape velocity from the surface of a body of mass M and radius R is $v_{\rm esc} = \sqrt{2GM/R}$. When M and R are such that this escape velocity $v_{\rm esc} \to c$, even light cannot escape from its surface (this is the line $M = (c^2/2G)R$). This is the historical definition of a "black hole" (Michell & Laplace, 18th century).

is Newton's gravitational constant. Near a Black Hole, $GM/Rc^2 \sim 1$, neutron star $\sim 10^{-1}$, near the sun (say at the orbital radius of mercury) it is $\sim 10^{-8}$ and $\sim 10^{-6}$ near the sun's surface, whereas near the earth's surface $GM/Rc^2 \sim 10^{-9}$. Note that near the earth and the sun, the precision of the measurements is sufficiently high that even for such small values of GM/Rc^2, we need to take deviations from Newtonian gravity into account. See Fig. 1.3 to get a feel for where different objects lie in the size-mass plane, and where how far they are from the $GM/Rc^2 \sim 1$

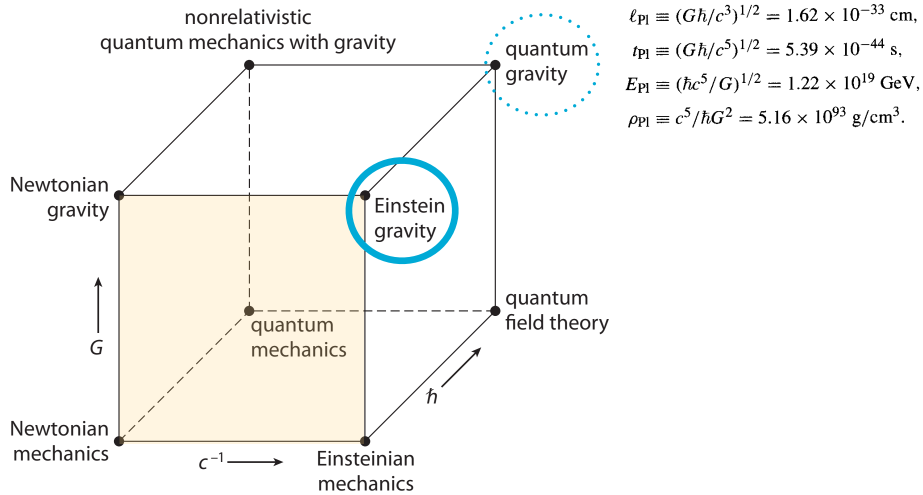

Figure 1.4: Cube of Physics, adapted from A. Zee.

region.

It is very useful to remember that for the sun $GM_\odot/c^2 \sim 1\,\text{km}$, where $M_\odot \sim 2 \times 10^{30}\,\text{kg}$.

Exercise 1.3.1 Estimate $GM/(Rc^2)$ for (i) a proton (ii) human (iii) moon (iv) earth (v) sun (vi) neutron star. Order of magnitude is fine.

1.4 What can GR do for you?

In the course, we will learn to appreciate (aspects of) the working of the Global Positioning System (GPS) relevant for your smart phones to help you orient yourself. Moving further out from our abode on our remarkable third rock from the sun, we will learn how the geometric understanding of spacetime explains the puzzling orbit of Mercury around the sun, and the observed deflection of light by the sun.[3] It also describes why we can occasionally see the same supernova go off multiple times – seeing the "echoes" of the same explosion. We will be able to describe the formation and environment of spectacular ultra-compact "objects", neutron stars and black holes in our cosmos. We will be able to understand how a dance of their binaries generate ripples in our dynamical spacetime, which have now been detected on our rock.[4] Time-permitting we will also touch upon how the evolution of the universe as a whole relies on our understanding of the laws governing a dynamical spacetime.

[3]Light deflection from the sun was first observed during a total solar eclipse of 1919. For a bit of history and its significance, you might like a talk I gave on this in the context of the solar eclipse that was viewable from the continental US in 2024. https://www.youtube.com/watch?v=D1LGSSoLfNg.

[4]See https://www.nobelprize.org/prizes/physics/2017/press-release/.

1.5 Cube of Physics

To conclude, let us also put general relativity in the context of classical and quantum mechanics. A nice visual way is via the "cube of physics", see Fig. 1.4. You should read this as, for example, when G becomes more relevant, then Newtonian gravity plays a prominent role. We will not have much to say about when quantum gravity in this course. It becomes important at the Big Bang and near the center of black holes. Notice the physical scale where quantum gravitational effects need to be taken into account. For this course, we will restrict ourselves to classical gravity.

CHAPTER 2

GEOMETRY AS PHYSICS

Gravity and Geometry Considerations of universal free-fall, amongst other things, eventually led Einstein to the profound insight that gravity is not a force; it is a manifestation of the *geometry of spacetime*. He was helped by the fact that his Special Theory of Relativity (ignoring gravitational effects), initially formulated in terms of transformation laws between different observers, had already been geometrized by Minkowski.[1] This geometry, whether in Special or General Relativity, is $3+1$ dimensional (3 spatial and 1 temporal dimension). In General Relativity, spacetime can be intrinsically curved, and this curvature encodes gravitational effects.

Geometry of space and spacetime is something accessible to experimental verification. Gauss could in principle have tested whether the angles of a large triangle sum to π (the triangle is in three dimensions, not restricted to be on the earth's surface).[2] Modern cosmological observations perform the same kind of experiment on vastly larger scales by comparing known physical scales with their observed angular sizes. You can learn about Gauss's failed attempt to do so in the vicinity of earth. And then learn about our very successful attempt to do so in the context of cosmology. Both are described in Chapter 2 of Hartle.

2.1 Euclidean vs. intrinsically curved spatial geometry

To understand "intrinsically curved" geometry, let us first restrict ourselves to spatial geometries (as opposed to spacetime geometry). To make things easier still, let us consider 2 dimensional spatial geometries.

2.1.1 Two dimensions

2d Euclidean An example of a Euclidean geometry is a 2-dimensional plane, where Euclid's axioms are satisfied. For example, (i) parallel lines do not intersect, (ii) sums of interior angles of

[1] Einstein initially resisted Minkowski's geometrization of Special Relativity, reportedly dismissing it as "superfluous learnedness." He is also quoted as saying, "Since the mathematicians have invaded the theory of relativity, I do not understand it myself anymore." The geometric viewpoint later became indispensable to Einstein in developing general relativity.

[2] This is historically debated a bit.

a triangle made of straight lines is π radians and (iii) the ratio of the circumference of a circle to its radius is 2π. Non-Euclidean geometry is of course a geometry where any of Euclid's axioms are violated.

2d intrinsically curved An example of an intrinsically curved 2-dimensional geometry is a 2-sphere (with radius a). Note that Euclid's axioms are not satisfied on a 2-sphere. For example: (i) Any two distinct great circles intersect, so there is no global analogue of distinct parallel straight lines on the sphere. (ii) Sums of interior angles of a triangle made of straight lines (great circles) is $> \pi$ radians ($= \pi + A/a^2$). Here, A is the area of the triangle on the 2-sphere. (iii) The ratio of the circumference of a circle to its radius is $< 2\pi$. That is, circumference(C)/radius (r)$= 2\pi \sin(r/a)/(r/a)$ where r is the radius of the circle on the 2-sphere.

Note that these properties can be tested by a bug living on the surface of the sphere, without ever leaving the surface. These are " intrinsic" properties of the surface, not requiring any information beyond the surface itself. For example, a bug could take a small segment of a string, and attach one end at some point on the sphere. Then crawl away till the string becomes taut. The bug would measure this taut length in terms of their own tiny steps. This is a radius r in units of their step size. Now, the bug can walk around keeping the string taut to trace out a circle on the sphere. Again, the circumference C can be measured in terms of the number of their tiny steps. By comparing the circumference to the radius, the bug would discover that they were living in a non-Euclidean space. They never had to see the embedded 2-sphere in 3d Euclidean space by flying out of the surface! Also note they can look for these effects without knowledge of global properties such as whether their space is finite or infinite.

A geometry is *intrinsically curved* if departures from Euclidean geometry can be detected by measurements made entirely within the space, even locally, without reference to an embedding in a higher-dimensional space. The opposite is *intrinsically flat.* Thus, a sphere is intrinsically curved, while a plane is intrinsically flat. A cylinder provides an instructive example: although it looks curved when embedded in three dimensions, it is intrinsically flat.

Intrinsically curved spatial geometries are necessarily non-Euclidean, although the term "non-Euclidean" by itself can also include global differences that do not involve intrinsic curvature.[3]

2.1.2 Three dimensions

3d Euclidean Our usual Euclidean 3d space is the obvious generalization of a plane from 2 to 3d.

3d intrinsically curved A 3-sphere is a 3d generalization of the 2-sphere. It is also non-Euclidean. For example the volume enclosed by a 2-sphere of radius r in this 3-sphere geometry is

$$V = 2\pi a^3 \left[\frac{r}{a} - \sin\left(\frac{r}{a}\right) \cos\left(\frac{r}{a}\right) \right] . \tag{2.1.1}$$

In the limit $r/a \ll 1$, this volume approaches its Euclidean counterpart $V = (4\pi r^3/3)[1 - (1/5)r^2/a^2 + \ldots]$.

[3] Euclidean geometries are intrinsically flat, but the converse need not hold globally. For example, an infinite cylinder is intrinsically flat and locally Euclidean, but is not globally equivalent to the Euclidean plane.

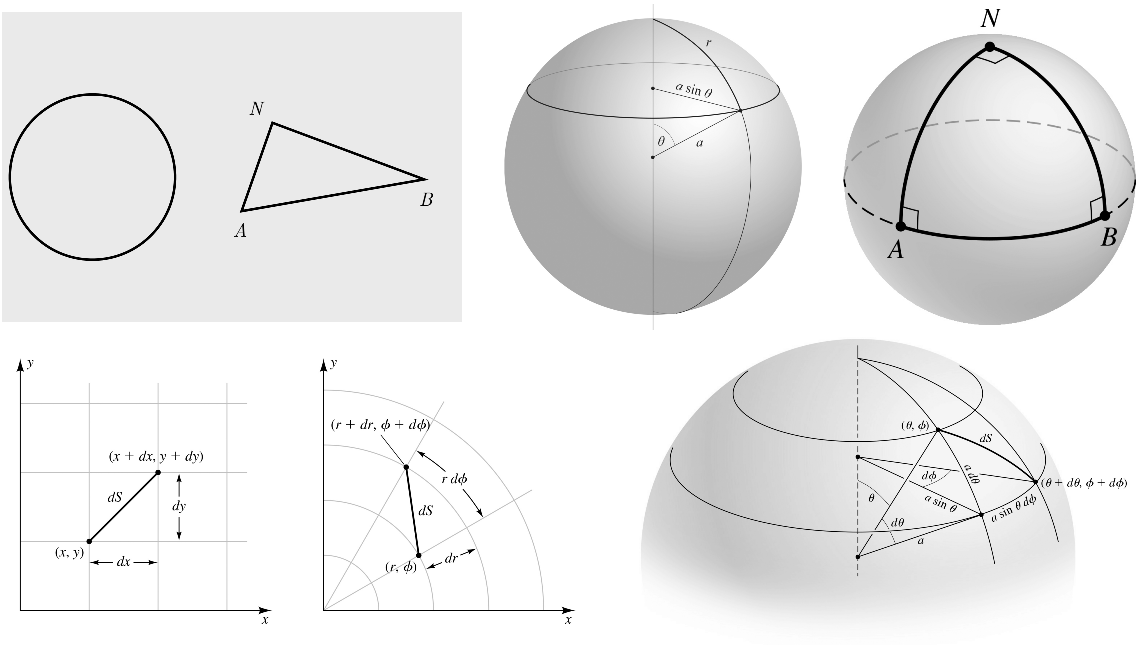

Figure 2.1: Top panel (right): The sum of the interior angles of a triangle ABN on a 2-sphere is $> \pi$, in this particular case it is $3\pi/2$. On the plane it is of course π. Top panel (middle sphere): The circumference of the circle of radius r is $< 2\pi r$, it is $C = 2\pi r \sin(r/a)/(r/a)$. On a plane it would be $C = 2\pi r$. Bottom panel (left): Cartesian and Polar co-ordinates in a 2d plane. Bottom panel (right): r and θ co-ordinates on the sphere. Part of the Fig. adapted from Hartle.

2.2 Specifying a geometry

How does one specify a geometry? A convenient and general way to do this is to define a rule for determining distances between nearby points. This is called a metric. Once this rule is specified, lengths of curves can be calculated by adding up such small distances. "Straight" lines can now be defined as curves with extremal lengths between two points. Circles (or spherical surfaces) can be defined as the locus of all points equidistant from a given point, angles can be defined as ratios of some arclength of a circle to the radius, and so on. This way of defining a geometry has significant advantages over trying to describe a geometry by embedding it in some higher dimensional Euclidean space or specifying it with a set of axioms. Both of these get cumbersome and difficult as we move to general geometries which do not have high degree of symmetry.

If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Please inform your Flat-Earther buddies.

2.2.1 Labeling and calculating with co-ordinates

Co-ordinates are labels of points in a geometry. For example, in a 2d plane, one can label points using the usual Cartesian co-ordinates (x, y). One could also label points using polar co-ordinates

(r, θ). Different co-ordinates can be chosen to label points in the same geometry. Different co-ordinate systems are connected by a co-ordinate transformation. For example, $x = r\cos\theta$ and $y = r\sin\theta$.

2.3 Spatial metric

The square of the distance, $\mathrm{d}S^2$, between nearby points in a plane with labels (x, y) and $(x + \mathrm{d}x, y + \mathrm{d}y)$ or (r, θ) and $(r + \mathrm{d}r, \theta + \mathrm{d}\theta)$

$$\mathrm{d}S^2 = \mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2. \tag{2.3.1}$$

In different co-ordinate systems the expression for the distance can look different, as can be seen above. Here is the metric in yet another co-ordinate system, $\mathrm{d}S^2 = 2\mathrm{d}\tilde{x}^2 + \mathrm{d}\tilde{y}^2 - 2\mathrm{d}\tilde{x}\mathrm{d}\tilde{y}$. This one also represents a 2d plane. The co-ordinates are linked by $(\tilde{x}, \tilde{y}) = (x, x + y)$. Changing co-ordinate systems does not change the geometry.

Exercise 2.3.1 Using $x = r\cos\theta$ and $y = r\sin\theta$, show that $\mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2$.

For a 3d Euclidean space, the distance between nearby points (x, y, z) and $(x + \mathrm{d}x, y + \mathrm{d}y, z + \mathrm{d}z)$ or equivalently, (r, θ, ϕ) and $(r + \mathrm{d}r, \theta + \mathrm{d}\theta, \phi + \mathrm{d}\phi)$ is

$$\mathrm{d}S^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = \mathrm{d}r^2 + r^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2). \tag{2.3.2}$$

The different co-ordinate systems are connected by $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ and $z = r\cos\theta$.

Similarly, for a non-Euclidean geometry like a 2-sphere of radius a, a convenient choice of co-ordinates are the latitudes and longitudes (θ, ϕ). The distance between nearby points (θ, ϕ) and $\theta + \mathrm{d}\theta, \phi + \mathrm{d}\phi)$ is

$$\mathrm{d}S^2 = a^2(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2). \tag{2.3.3}$$

For a 3-sphere of radius a, the distance between nearby points (ψ, θ, ϕ) and $(\psi + \mathrm{d}\psi, \theta + \mathrm{d}\theta, \phi + \mathrm{d}\phi)$ is

$$\mathrm{d}S^2 = a^2[\mathrm{d}\psi^2 + \sin^2\psi(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2)]. \tag{2.3.4}$$

While not necessary, you can embed this 3-sphere ib 4d Euclidean space as a 3-surface $x^2 + y^2 + z^2 + w^2 = a^2$.

Note that this square of small distances can be conveniently written as

$$\mathrm{d}S^2 = \sum_{i=1}^{3}\sum_{j=1}^{3} g_{ij}\mathrm{d}x^i\mathrm{d}x^j, \tag{2.3.5}$$

where $x^1 = \psi$, $x^2 = \theta$ and $x^3 = \phi$, where $g_{11} = a^2$, $g_{22} = a^2\sin^2\psi$, $g_{33} = a^2\sin^2\psi\sin^2\theta$ with all other $g_{ij} = 0$ (the off-diagonal terms need not always be zero). We will use a simplified notation

$$\mathrm{d}S^2 = g_{ij}\mathrm{d}x^i\mathrm{d}x^j, \tag{2.3.6}$$

where summation over repeated upstairs and down stairs indices is implied.[4] Note that g_{ij} is a function of the co-ordinates $g_{ij}(x^1, x^2, x^3)$, which we will write as $g_{ij}(x^k)$ in shorthand.

For a given co-ordinate system, you can think of g_{ij} as components of a $d \times d$ matrix $\boldsymbol{g}$ where d is the dimension of space, with g_{ij} being the entry on the i-th row and j-th column. These entries will depend on the co-ordinate system. For example, in the case of the metric in 3d Euclidean space, the matrix entries in Cartesian co-ordinates would be $g_{ij} = \delta_{ij}$ where $\delta_{ij} = 1$ for $i = j$ and 0 for $i \neq j$. In Polar co-ordinates, $g_{11} = 1$, $g_{22} = r^2$ and $g_{33} = r^2 \sin^2\theta$, and all other entries being 0. The components (matrix entries) look different, but together with $\mathrm{d}x^i \mathrm{d}x^j$ in the respective co-ordinates, they must yield the same $\mathrm{d}S^2$.

In general, the infinitesimal, $\mathrm{d}S^2 = g_{ij}\mathrm{d}x^i\mathrm{d}x^j$, that determines the distance between any two nearby points is called the metric. In the case, since we are restricting ourselves to space and not spacetime, it is the spatial metric. It is also commom practice to call g_{ij} the metric also.

Exercise 2.3.2 (i) Write down the components g_{ij} on a 2-sphere of radius a in terms of θ and ϕ co-ordinates. (ii) Consider the "Pinched Peanut" geometry defined by $\mathrm{d}S^2 = a^2(\mathrm{d}\theta^2 + f^2(\theta)\mathrm{d}\phi^2)$ where $f(\theta) = \sin\theta(1 - \sin^2\theta)$. Write down the components g_{ij} in terms of θ and ϕ co-ordinates.

2.3.1 Path length and volumes

Path length : To calculate non-infinitesimal distances along some curve γ, we can use the metric (in any co-ordinate system) as follows:

$$S[\gamma] = \int_\gamma \mathrm{d}S = \int_{\lambda_1}^{\lambda_2} \mathrm{d}\lambda \sqrt{g_{ij}\frac{\mathrm{d}x^i}{\mathrm{d}\lambda}\frac{\mathrm{d}x^j}{\mathrm{d}\lambda}}. \tag{2.3.7}$$

In the final expression, you should think of a parametrized curve $x^i(\lambda)$, and $g_{ij}(x^k(\lambda))$. The length of the curve is of course independent of the choice of co-ordinates, so it can be computed in any co-ordinate system.

Suppose Bob the bug is walking on a 2-sphere, leaving a localized chemical trail. Will the length of this chemical trail depend on the co-ordinates or parametrization we use to characterize the curve. Of course not! Nevertheless, depending on the geometry of space and the curve in question, it is useful to make a good co-ordinate choice for computational ease. It might be also useful sometimes to use one of the appropriately chosen co-ordinates itself as the parameter λ.

Important: Geometric quantities, for example, the distance between two nearby points, or the length of a particular curve, does not depend on the choice of co-ordinate systems or parametrization.

Exercise 2.3.3 Consider a 2d plane. A parabola passing through the origin can be parametrized as $x^1(\lambda) = \lambda^2/2$ and $x^2(\lambda) = \lambda$, where x^1 and x^2 are Cartesian co-ordinates. Write down g_{ij} and

[4] Notice the potentially confusing notation, $\mathrm{d}x^2 \neq (\mathrm{d}x)^2$ in these later expressions, although earlier we were using $\mathrm{d}x^2 = (\mathrm{d}x)^2$.

$\mathrm{d}x^i(\lambda)/\mathrm{d}\lambda$. Compute the length between $\lambda = 0$ and $\lambda = 1$. Hint: You might find the following substitution $\lambda = \sinh u$ useful for doing the integration.

Volumes To calculate the volume of a spatial region enclosed by a surface $\mathcal{S}$

$$\mathcal{V} = \int_{\mathcal{V}} \sqrt{\det g} \prod_{i=1}^{d} \mathrm{d}x^i, \tag{2.3.8}$$

where $\det g$ is the determinant of $\boldsymbol{g}$. To get some intuition, consider a metric which is diagonal in some co-ordinate system ($g_{ij} = 0$ for $i \neq j$). Then the infinitesimal distance along each co-ordinate direction when the co-ordinates change by $\mathrm{d}x^i$ is $\mathrm{d}S_i = \sqrt{g_{ii}}\mathrm{d}x^i$ (no summation). And the volume will be $\prod_{i=1}^{d} \mathrm{d}S_i$. Note that the diagonality of g_{ij} means the co-ordinate curve intersect at right angles.

Exercise 2.3.4 Use the metric on the surface of a 2-sphere to do the following: (i) Calculate the circumference a circle of radius r (distance from the center to the circumference) in the two-dimensional geometry given by the surface of a sphere of radius a. Show that this reduces to $2\pi r$ when $r \ll a$. (ii) Calculate the area of the same circle on the sphere, and again show that it reduces to πr^2 when $r \ll a$. Bob, the bug lives here. (iii) Do the same for a cylinder of radius a and infinite extent in the perpendicular direction to the radius. Assume $r < \pi a$. Note that the cylinder is not globally Euclidean, but it is intrinsically flat.

Exercise 2.3.5 Beetle Bina lives in a different two-dimensional geometry with metric

$$\mathrm{d}S^2 = \mathrm{d}r^2 + a^2 \sinh^2(r/a)\, \mathrm{d}\phi^2. \tag{2.3.9}$$

Here $0 \leq \phi < 2\pi$, and r is the intrinsic radial distance from $r = 0$. (i) Calculate the circumference $C(r)$ of a circle of radius r. (ii) Calculate the area $A(r)$ enclosed by this circle. (iii) Show that for $r \ll a$,

$$C(r) = 2\pi r \left[1 + \frac{r^2}{6a^2} + \cdots\right], \tag{2.3.10}$$

and

$$A(r) = \pi r^2 \left[1 + \frac{r^2}{12a^2} + \cdots\right]. \tag{2.3.11}$$

Compare these results with those for a Euclidean plane. For a circle of fixed radius r, does Bina have more area enclosed or does Bob? (See the previous problem for Bob).

Having hopefully gained some intuition about what intrinsically curved means, let me re-iterate that while we started with just curved space for pedagogy, ultimately we care about curved spacetime when thinking about gravitational physics.

For spacetime, the role played by Euclidean spatial geometry in the examples in this chapter, will be taken over by Minkowski spacetime geometry (which is non-Euclidean!).[5] And, we will define instrincally curved spacetime geometry by their departure from Minkowski spacetime geometry.

2.4 Practice Problems and Beyond

1. **Triangles on a Sphere**: Draw the following triangles on a 2-sphere (each side is part of a great circle). (i) The sum of interior angles is almost π. (ii) The sum of interior angles is almost 2π.

2. **Arc Lengths**: Using $\mathrm{d}S = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}$ show that the perimeter enclosing a half circle (the arc + straight line) of radius a, $C = \oint \mathrm{d}S = \oint \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = \pi a + 2a$. Similarly, use $\mathrm{d}S = \sqrt{\mathrm{d}r^2 + r^2\mathrm{d}\theta^2}$ in polar co-ordinates to find C also.

3. **More Peanuts**: Consider the peanut geometry described by the metric $\mathrm{d}S^2 = a^2(\mathrm{d}\theta^2 + f^2(\theta)\mathrm{d}\phi^2)$ where $f(\theta) = \sin\theta(1 - 3\sin^2\theta/4)$. For $\theta = \Theta < \pi/2$ in this geometry, calculate the radius of the cap (you can't leave the surface again) enclosed by the circle defined by $\theta = \Theta$, and how the circumference of this circle is connected to the radius.

4. **Spherical Geometry 2d** Consider a latitude where $\theta = \Theta < \pi/2$ on a 2-sphere of radius a. Show that the radius of the spherical cap (you can't leave the surface!) enclosed by this latitude is $r = a\Theta$, the length of the latitude is $2\pi a \sin\Theta$. A useful generalization, if you want to know the distance between any two points on such a sphere (without restricting to the same latitude or longitude is as follows: The distance between two points on a 2-sphere or radius a, with co-ordinates (θ_1, ϕ_1) and (θ_2, ϕ_2) is

$$S = a\cos^{-1}\left[\sin\theta_1\sin\theta_2 + \cos\theta_1\cos\theta_2\cos(\phi_1 - \phi_2)\right]. \tag{2.4.1}$$

   See, for example, here for details.

5. **Not-so-flat Earth**: If the distances along the surface between all nearby cities of the earth were given, you would have been able to tell that the earth is not a plane. Indeed, if someone told you all the distances between any 4 four non-collinear cities, you would know that the surface of the earth is not planar. Let P_1, P_2, P_3 and P_4 be the four cities, with $S_{ij} = S_{ji}$ being the distance between them (length of the shortest curve in the surface between those points). Then, the 4-points lie in a plane if and only if:

$$\det\begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & S_{12}^2 & S_{13}^2 & S_{14}^2 \\ 1 & S_{21}^2 & 0 & S_{23}^2 & S_{24}^2 \\ 1 & S_{31}^2 & S_{32}^2 & 0 & S_{34}^2 \\ 1 & S_{41}^2 & S_{42}^2 & S_{43}^2 & 0 \end{pmatrix} = 0. \tag{2.4.2}$$

[5]The metric (in spatially Cartesian co-ordinates) is

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2. \tag{2.3.12}$$

It is the negative sign that makes it non-Euclidean.

Check this, and then please inform your flat-earther buddies. If you want to confirm that we are on an (idealized) 2-sphere, then the following must be true:

$$\det \begin{pmatrix} 1 & \cos S_{12} & \cos S_{13} & \cos S_{14} \\ \cos S_{21} & 1 & \cos S_{23} & \cos S_{33} \\ \cos S_{31} & \cos S_{32} & 1 & \cos S_{34} \\ \cos S_{41} & \cos S_{42} & \cos S_{43} & 1 \end{pmatrix} = 0. \tag{2.4.3}$$

For proofs, and more, see this post by Terence Tao here.

2.5 Appendix

2.5.1 Curves of Extremal Length

The expression for $S[\gamma]$ is the length of any given curve γ:

$$S[\gamma] = \int_\gamma \mathrm{d}S = \int_{\lambda_1}^{\lambda_2} \mathrm{d}\lambda \sqrt{g_{ij} \frac{\mathrm{d}x^i}{\mathrm{d}\lambda} \frac{\mathrm{d}x^j}{\mathrm{d}\lambda}}. \tag{2.5.1}$$

Question: Given some points P and Q in our geometry, I want to find the curve connecting those points such that its length is a minimum (actually just "stationary"). The answer is provided by the curve $x^i(\lambda)$ which satisfies the **Euler-Lagrange equations**:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial(\mathrm{d}x^i/\mathrm{d}\lambda)}\right) - \frac{\partial L}{\partial x^i} = 0, \qquad \text{where} \qquad L = \sqrt{g_{ij} \frac{\mathrm{d}x^i}{\mathrm{d}\lambda} \frac{\mathrm{d}x^j}{\mathrm{d}\lambda}}. \tag{2.5.2}$$

Curves satisfying this equation have the property that between two fixed end points, if I try to change the curve γ to $\gamma + \delta\gamma$, the length of the curve does not change at first order in the small changes $\delta S[\gamma] = S[\gamma + \delta\gamma] - S[\gamma] = 0 + \mathcal{O}[\delta\gamma^2]$. The analogues of straight lines in arbitrary spatial geometries are called *geodesics*. They are curves whose length is stationary under sufficiently small variations with fixed endpoints, the ones we just found above.

Straight Line in a plane: Let me show you the use of this "technology" to go through a long route to find the equation of a straight line in a 2d plane. While seemingly trivial, understanding how it is used here will help you gain confidence in using it in more complicated situations later. I will also introduce the (sometimes confusing) notion of an affine parameter.

Consider two points P and Q in a plane. Let us put Cartesian co-ordinates (x, y) on the plane. Consider a collection of curves with P and Q as their endpoints. Each curve is parametrized by λ: $(x(\lambda), y(\lambda))$. We move along a curve from P to Q as λ changes. We wish to find the curve connecting P and Q with extremal length. Of course you intuitively know that this must be a straight line, ie. $\mathrm{d}y/\mathrm{d}x = const$. We will prove this. In terms of the parameterized curve, you perhaps might also be expecting the equation that the curve satisfies is $\mathrm{d}^2x(\lambda)/\mathrm{d}\lambda^2 = 0$ and $\mathrm{d}^2y(\lambda)/\mathrm{d}\lambda^2 = 0$. We will see that this will not be true in general for arbitrary choice of λ, but will turn out to be true a special class of them. This special class is: $\lambda = aS + b$ where S is the arc length along the curve for arbitrary choice of $a \neq 0$ and b.

Let us begin. The length of any curve γ from P to Q

$$\begin{aligned} S[\gamma] = \int_\gamma \mathrm{d}S = \int \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} &= \int \mathrm{d}\lambda \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\lambda}\right)^2}, \\ &= \int \mathrm{d}\lambda\, L, \quad \text{where} \quad L = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\lambda}\right)^2}. \end{aligned} \tag{2.5.3}$$

If a curve extremizes its length, then $\delta S = 0$. Such a curve satisfies the Euler-Lagrange equations

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial(\mathrm{d}x/\mathrm{d}\lambda)}\right) = \frac{\partial L}{\partial x}, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\partial L}{\partial(\mathrm{d}y/\mathrm{d}\lambda)}\right) = \frac{\partial L}{\partial y}. \tag{2.5.4}$$

These equations yield:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right) = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}y}{\mathrm{d}\lambda}\right) = 0. \tag{2.5.5}$$

Since these equations imply $L^{-1}\mathrm{d}x/\mathrm{d}\lambda = const.$ and $L^{-1}\mathrm{d}y/\mathrm{d}\lambda = const.$, taking the ratio of these constants is also a constant. That is $\mathrm{d}y/\mathrm{d}x = const.$ This is of course the expected equation for the locus of a line.

However, these equations do not yield $\mathrm{d}^2x/\mathrm{d}\lambda^2 = 0$ and $\mathrm{d}^2y/\mathrm{d}\lambda^2 = 0$. Now, look back at the equation for S. You will notice that $\mathrm{d}\lambda L = \mathrm{d}S$, where $\mathrm{d}S$ is the infinitesimal length along the curve. So our equation for $x(\lambda)$ can be written as (after multiplying by L^{-1})

$$\frac{1}{L}\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right) = 0 \implies \frac{\mathrm{d}^2x}{\mathrm{d}S^2} = 0, \tag{2.5.6}$$

where we used $L\mathrm{d}\lambda = \mathrm{d}S$. Similarly for the y equation.

Are there a general class of parameters λ' for which $\mathrm{d}^2x/\mathrm{d}\lambda'^2 = 0$? Clearly if $\mathrm{d}S = \mathrm{d}\lambda'/a$ or equivalently, if $\lambda' = aS + b$, then again $\mathrm{d}^2x/\mathrm{d}\lambda'^2 = 0$. Similarly for y. Thus all parameters λ' which are related linearly to the distance along the curve S will yield the desired simple equations. These are called **affine parameters**.

Note that regardless of the parametrization, the locus is $\mathrm{d}y/\mathrm{d}x = const.$ – it is a line as you expect. However $\mathrm{d}^2x/\mathrm{d}\lambda^2 = 0$ and $\mathrm{d}^2y/\mathrm{d}\lambda^2 = 0$ for along this straight line locus of points only if we chose λ to be an affine parameter.

CHAPTER 3

FROM NEWTONIAN SPACE AND TIME TO MINKOWSKI SPACETIME

I am combining aspects of Ch. 3 and Ch. 4 of Hartle here.

3.1 Newtonian Space, Time and Reference Frames

The arena of Newtonian physics is a 3-dimensional Euclidean space and a universal time. However, this seemingly obvious (from everyday experience at least) Newtonian separation into an absolute Euclidean space plus a universal time will be replaced by a "spacetime" by the end of this chapter. To get there, we will first need to understand the notion of inertial reference frames, Galilean and Lorentz transformations between such frames, as well as the Principle of Relativity. I hope to make each step seem obvious, and yet the conclusion is anything but obvious.

Reference Frames In Euclidean space, imagine an idealized 3d rigid lattice of rods. At each intersection there is an idealized clock, with all ticking at some universal rate. I like to think of tiny demons sitting at these intersections, meticulously recording whatever happens in their vicinity. This lattice of rods and clocks defines a *reference frame*. It allows us to label where and when events take place.

Although the lattice naturally suggests Cartesian co-ordinates, we could equally well use spherical polar co-ordinates, or some other choice. How we label points is a choice; the reference frame is the same. Furthermore, not just co-ordinates of a given reference frame, but the choice of reference frame itself does not change the underlying (assumed Euclidean) geometry of space, and universality of time.

Different reference frames can be displaced, rotated, rotating, or moving with respect to each other. Among this large family, there is a special class of frames called *inertial frames*. Within the Newtonian space and time context, we define them below.

3.1.1 Inertial Reference Frames

Go far away from large masses and shield yourself, as well as you can, from other long-range forces. Set up a reference frame with Cartesian co-ordinates (x, y, z) and synchronized clocks. If a free particle moving through the frame satisfies

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = \frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = 0, \tag{3.1.1}$$

then your frame is *inertial*. Any frame moving with constant velocity with respect to an inertial frame is also inertial. A rotated (but not rotating) or displaced version of an inertial frame is also inertial.

3.2 Galilean Transformations

Consider two inertial frames $\mathcal{S}$ (Sammy's frame) and $\mathcal{S}'$ (prime is Polly's frame) in the standard configuration: their axes are aligned, their origins coincide at $t = t' = 0$, and $\mathcal{S}'$ moves with speed v along the x-axis of $\mathcal{S}$. Then, the relationships between the co-ordinates and time of an event is give by

$$x = x' + vt', \qquad y = y', \qquad z = z', \qquad t = t'. \tag{3.2.1}$$

For a particle with velocity $(V^{x'}, V^{y'}, V^{z'})$ in $\mathcal{S}'$,

$$(V^x, V^y, V^z) = (V^{x'} + v, V^{y'}, V^{z'}). \tag{3.2.2}$$

Notice that $\mathrm{d}^2 x/\mathrm{d}t^2 = 0 \Longrightarrow \mathrm{d}^2 x'/\mathrm{d}t'^2 = 0$ and similarly for y' and z'.

3.3 Principle of Relativity

No experiment performed entirely inside an inertial laboratory can distinguish one inertial frame from another.

Consider two inertial frames, which we will imagine to be enclosed opaque boxes (with grids and clocks inside them as before), with our previous $\mathcal{S}$ being Sammy's frame and $\mathcal{S}'$ being Polly's. The Principle of Relativity states that if Sammy and Polly set up exactly the same initial configurations of matter and fields in their respective inertial laboratories, the subsequent evolution of the matter and fields will be identical in their respective frames. That is, the laws of physics are the same in all inertial frames.

Galileo, said it best (from his *Dialogue Concerning the Two Chief World Systems*):

> Shut yourself up with some friend in the main cabin below decks on some large ship, and have with you there some flies, butterflies, and other small flying animals. Have a large bowl of water with some fish in it; hang up a bottle that empties drop by drop into a wide vessel beneath it. With the ship standing still, observe carefully how the little animals fly with equal speed to all sides of the cabin. The fish swim indifferently in all directions; the drops fall into the vessel beneath; [...] When you have observed all these things carefully [...], have the ship proceed with any speed you like, so long as the motion is uniform and not fluctuating this way and that. You will discover not the least change in all the effects named, nor could you tell from any of them whether the ship was moving or standing still. In throwing something to your companion, you will need no more force to get it to him whether he is in the direction of the

> bow or the stern, with yourself situated opposite. [...] the butterflies and flies will continue their flights indifferently toward every side, nor will it ever happen that they are concentrated toward the stern, as if tired out from keeping up with the course of the ship, from which they will have been separated during long intervals by keeping themselves in the air.

Note that Galileo was talking about mechanical phenomenon, our principle, as used by Einstein applies to all phenomenon.

3.3.1 "Light" Trouble for Galilean Transformations

If Sammy and Polly set up identical experiments with a laser in their respective laboratories and measure the speed of light, they both find $c = 3 \times 10^{10}\,\mathrm{cm\,s^{-1}}$. This is consistent with the Principle of Relativity (PoR).

Now let them drill holes in their laboratories so that light emitted by Polly's laser enters Sammy's laboratory. Sammy measures the speed of this incoming light and again finds it to be c.[1] Note that this is a different statement from the previous one: before, Sammy and Polly performed identical experiments within their own laboratories; now they are describing the same physical light sent from Polly's laboratory to Sammy's.

This spells trouble for Galilean Transformations (GT). Suppose Polly's reference frame moves with velocity $v\hat{\boldsymbol{x}}$ relative to Sammy's, and Polly sends the light toward Sammy in the $-\hat{\boldsymbol{x}}$ direction. Polly measures the velocity of the light to be $-c\,\hat{\boldsymbol{x}}$. According to GT, Sammy should therefore measures its velocity to be $(v-c)\hat{\boldsymbol{x}}$, and hence its speed should be $|v-c|$. But Sammy instead measures the speed of the light to be exactly $c = 3 \times 10^{10}\,\mathrm{cm\,s^{-1}}$, independent of the relative velocity v between the two laboratories.

The prediction $|v-c|$ arose directly from GT. If we retain the Principle of Relativity and the observed behavior of light, GT must therefore be wrong.

3.4 Resolution via Lorentz Transformations

We can leave PoR as is, but change GT as follows. Einstein suggested the transformation between frames should be[2]

$$x = \gamma(x' + vt'), \quad y = y', \quad z = z', \quad t = \gamma(t' + vx'/c^2), \tag{3.4.1}$$

where

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{3.4.2}$$

With these transformations, the velocities are related by

$$(V^x, V^y, V^z) = \left(\frac{V^{x'} + v}{1 + vV^{x'}/c^2}, \frac{V^{y'}}{\gamma(1 + vV^{x'}/c^2)}, \frac{V^{z'}}{\gamma(1 + vV^{x'}/c^2)} \right). \tag{3.4.3}$$

[1] This is what Michelson-Morley essentially found, though they were really looking for motion relative to ether.

[2] They carry Lorentz name because he had already discovered these transformations in the context of "ether" theory. Einstein gave them a fundamentally different interpretation.

When $v/c \ll 1$, this reduces to the Galilean result of $V^x \approx V^{x'} + v$ and so on. But if $V^{x'} = \pm c$, then

$$V^x = \pm c, \tag{3.4.4}$$

independent of v.

3.5 The "Price"

We have retained the Principle of Relativity and the observed behavior of light, but replaced Galilean Transformations by Lorentz Transformations. There is a price: the Newtonian notions of a universal time and a frame-independent spatial length can no longer be maintained.

Now? : Recall that under Galilean Transformations, $x = x' + vt', t = t'$. In particular, if two events occur at the same time for Polly, $\mathrm{d}t' = 0$, then they also occur at the same time for Sammy, $\mathrm{d}t = 0$.

Now, recall that with Lorentz transformations $x = \gamma(x' + vt')$ and $t = \gamma\left(t' + vx'/c^2\right)$. For two nearby events,

$$\mathrm{d}t = \gamma\left(\mathrm{d}t' + \frac{v}{c^2}\mathrm{d}x'\right). \tag{3.5.1}$$

Therefore, if Polly regards the two events as simultaneous, $\mathrm{d}t' = 0$, then Sammy finds

$$\mathrm{d}t = \frac{v}{c^2}\mathrm{d}x' \neq 0, \tag{3.5.2}$$

for spatially separated events ($\mathrm{d}x' \neq 0$).

Two events that are simultaneous for Polly need not be simultaneous for Sammy. There is no universal notion of "now."

Length? : This also changes our notion of lengths. Measuring the length of an extended object requires determining the positions of its two ends *simultaneously*, that is, at the same time. Since Sammy and Polly no longer agree on which pairs of events are simultaneous, they need not agree on the spatial length of an object. Spatial separation at a fixed time, like time intervals, can depend on the inertial frame.

3.6 Invariant spacetime interval and Minkowski spacetime

What replaces these separately observer-independent notions of spatial separations and time intervals is a frame-independent quantity associated with *spacetime*. Define

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 \tag{3.6.1}$$

Using Lorentz Transformations, one finds

$$-c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = -c^2\mathrm{d}t'^2 + \mathrm{d}x'^2 + \mathrm{d}y'^2 + \mathrm{d}z'^2. \tag{3.6.2}$$

Thus, for the same events, Sammy and Polly might find $\mathrm{d}x \neq \mathrm{d}x'$ and $\mathrm{d}t \neq \mathrm{d}t'$, but they agree on the spacetime interval $\mathrm{d}s^2$.

Exercise 3.6.1 For the standard configuration of reference frames, show that $-c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 = -c^2\mathrm{d}t'^2 + \mathrm{d}x'^2 + \mathrm{d}y'^2 + \mathrm{d}z'^2$.

The arena of physics is therefore no longer naturally separated into a universal 3-dimensional Euclidean space, and a universal time. Space and time must instead be regarded as parts of a single 3+1 dimensional spacetime.

Just like the geometry of space was specified by providing $\mathrm{d}S^2$ between any two points in that space, the geometry of the spacetime we have arrived at is defined by the spacetime metric

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2, \tag{3.6.3}$$

where $\mathrm{d}S^2$ is the Euclidean metric.

Here, we have chosen the synchronized time coordinate t of the inertial frame (see below), while allowing arbitrary time-independent coordinates on its Euclidean spatial slices; $\mathrm{d}S^2$ is the Euclidean spatial line element written in whatever spatial coordinates we choose. The geometry defined by $\mathrm{d}s^2$ is the **Minkowski spacetime**. Notice the minus sign. Minkowski spacetime is not Euclidean. But it is not intrinsically curved either.

Inertial Reference frames in Minkowski spacetime We can again imagine a lattice of rods and clocks, with the same procedure of choosing an intertial frame.

The clocks now need to be *synchronized* carefully. A convenient operational prescription is Einstein synchronization: if one clock sends a light signal at t_e, a nearby clock reflects it, and the first receives it at t_r, then the nearby clock is assigned the reflection time $t_{\text{bounce}} = \frac{1}{2}(t_e + t_r)$.

The care is necessary, because if we move clocks from a common location where they would be synchronized, the time they show when they get to their eventual grid point can depend on how we moved them! We will see this by the end of this chapter when we discuss proper time.

If we move from one inertial frame to another, but in each inertial frame we adopt a synchronized time, the (right hand side) of the metric can be written as $-c^2\mathrm{d}t^2 + \mathrm{d}S^2$ where $\mathrm{d}S^2$ is the metric of Euclidean space. We could also label the frame using some co-ordinates that mix space and time, in that case the rhs form will look different.

Exercise 3.6.2 For a given inertial reference frame, consider labeling the Euclidean space with Cartesian co-ordinates, then $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2$. (i) If instead you used polar co-ordinates $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$ and $z = r\cos\theta$ to label the Euclidean space, what would the right hand side be? (ii) Now consider labeling it with $x = \tilde{x}$ and $t = \tilde{t} - \alpha\tilde{x}/c$, $y = \tilde{y}$ and $z = \tilde{z}$ (with $|\alpha| < 1$). What is the form of the metric? Note that this is still the same underlying geometry of Minkowski space.

3.6.1 The Minkowski Metric

In general, the Minkowski metric (for that matter, any spacetime metric) can be written as

$$\mathrm{d}s^2 = g_{\mu\nu}(x^\alpha)\mathrm{d}x^\mu\mathrm{d}x^\nu, \tag{3.6.4}$$

where μ and ν range from 0 to 3 and Einstein summation convention is used over repeated greek indices. Here, $g_{\mu\nu}$ can be thought of as entries of a 4 symmetric matrix. In general its entries can depend on space and time co-ordinates (x^0, x^1, x^2, x^3). We used $g_{\mu\nu}(x^\alpha)$ as a shorthand for the $g_{\mu\nu}(x^0, x^1, x^2, x^3)$. For Minkowski space, if at each fixed-Einstein synchronized-time-slice within an inertial reference frame (with $x^0 = ct$), we choose to label the spatial slice with Cartesian co-ordinates then

$$g_{\mu\nu}(x^\alpha) \equiv \eta_{\mu\nu}, \tag{3.6.5}$$

where

$$\eta_{00} = -1, \quad \eta_{ij} = \delta_{ij}, \quad \eta_{0i} = 0. \tag{3.6.6}$$

When we think of $\eta_{\mu\nu}$ as entries of a matrix $\boldsymbol{\eta}$, then

$$\boldsymbol{\eta} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.6.7}$$

The entries of the inverse of this matrix are denoted by $\eta^{\mu\nu}$. Note that $\eta^{\mu\alpha}\eta_{\alpha\nu} = \delta^\mu{}_\nu$, where $\delta^0{}_0 = \delta^1{}_1 = \delta^2{}_2 = \delta^3{}_3 = 1$ and 0 for $\mu \neq \nu$. More generally, for any metric, we define $g^{\mu\nu}$ as entries of the inverse of the metric matrix:

$$g^{\mu\alpha} g_{\alpha\nu} = \delta^\mu{}_\nu \,. \tag{3.6.8}$$

Exercise 3.6.3 In an inertial frame in an underlying Minkowski spacetime we choose spacetime co-ordinates such $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2$, where $\mathrm{d}S^2$ is the metric on Euclidean spacetime. If we put spherical polar co-ordinates on the Euclidean space, what are $g_{\mu\nu}$ and $g^{\mu\nu}$ for all μ and ν. Note that μ and ν range from 0 to 3.

3.7 "Moving" in Spacetime

The best way to explore Minkowski spacetime is to think about the paths of particles through it.

For two nearby events:

$$\mathrm{d}s^2 < 0 \qquad \text{timelike,} \tag{3.7.1}$$
$$\mathrm{d}s^2 = 0 \qquad \text{lightlike or null,} \tag{3.7.2}$$
$$\mathrm{d}s^2 > 0 \qquad \text{spacelike.} \tag{3.7.3}$$

The locus of particles (massive or massless) in spacetime are called world lines. Massive particles move on timelike worldlines – that is $\mathrm{d}s^2 < 0$ along every segment of the worldline. On the other hand, light moves on null worldlines with $\mathrm{d}s^2 = 0$.

3.7.1 Spacetime Diagrams

Spacetime diagrams are invaluable for keeping track of events and comparing different frames. Always label the events you are comparing. For simplicity, consider $1+1$ dimensions, and we will be in an inertial frames with co-ordinates chosen so that $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 = -c^2\mathrm{d}t'^2 + \mathrm{d}x'^2$. We draw ct and x on an ordinary sheet of paper, but remember that the geometry of the diagram is not Euclidean.[3] Two distinct events can even satisfy $\mathrm{d}s^2 = 0$, something impossible for Euclidean distance.

Exercise 3.7.1 (i) Draw the wordline of a massive particle at rest $x = x_0$ in the ct vs. x plane. (ii) Draw the wordline of a free massive paricle, and a free massless particle starting at $x = x_0$ at $t = 0$ and moving in the negative x direction. (iii) Draw the wordline of a massive particle undergoing simple harmonic motion around $x = x_1$. (iv) Draw the following event. A very massive particle at rest at $x = 0$ emits two photons in the positive and negative x direction. The photon bounces off mirrors at $x = x_1$ and $x = -x_1$ and return back to the massive particle. Drawn wordlines of all three particles.

3.7.2 Two Frames in the same spacetime diagram

In the standard configuration, for a frame $\mathcal{S}'$ moving at speed v along x relative to $\mathcal{S}$:

Time axis of $\mathcal{S}'$ The origin of $\mathcal{S}'$ satisfies $x' = 0$, so

$$x = vt \quad \text{equivalently} \quad ct = \frac{c}{v}x. \tag{3.7.4}$$

Space axis of $\mathcal{S}'$ Events with $t' = 0$ satisfy

$$ct = \frac{v}{c}x. \tag{3.7.5}$$

These tilted axes already show that simultaneity is frame-dependent. It is very useful to make a grid of constant x' and t' lines in the ct vs. x plane.

Using spacetime diagrams, let us visit time dilation and length contraction:

Time dilation For a clock at rest in $\mathcal{S}'$, $\Delta x' = 0$, so invariance of the interval gives

$$c^2\Delta t'^2 = c^2\Delta t^2 - \Delta x^2. \tag{3.7.6}$$

Since $\Delta x = v\Delta t$,

$$\Delta t = \gamma \Delta t'. \tag{3.7.7}$$

A clock moving relative to us accumulates *less* time between the same two events $\Delta t' < \Delta t$.

[3] This is similar to saying that we can represent the geometry of a 2-sphere (our earth) on a flat sheet using a Mercater projection. The distance between two points on the Mercater projection however is not given by the Euclidean distance between those points!

Length contraction A rod of length L' is at rest in the $\mathcal{S}'$ frame. The length is measure at the same instant of time t' in the $\mathcal{S}'$ frame. We want to know the length in the $\mathcal{S}$ frame, which we can do, by recording the positions of its endpoints at the same time in that frame. The result is

$$L = \frac{L'}{\gamma}. \tag{3.7.8}$$

Be careful: length measurements in two different frames generally involve different pairs of simultaneous events.

Here are some details: In the $\mathcal{S}'$ frame, the end points of the rod are at $x' = 0$ and $x' = L'$. The equation of world line of these endpoints in the ct vs. x plane in the $\mathcal{S}$ frame are $ct = (c/v)x$ and $ct = (c/v)(x - L)$, where L is the length to be determined. Look at where these worldlines curves intersect the $t = 0$ axes, the interval on the x axis is L (we can of course do this at any fixed t, but $t = 0$ is convenient). Using $t = \gamma(t' + vx'/c^2)$, the farther endpoint's t' co-ordinate at $t = 0$, $t' = -L'v/c^2$, which can now be used in $x = \gamma(x' + vt')$ again for the farther endpoint to yield $L = \gamma(L' - v^2L'/c^2) = L'/\gamma$.

Exercise 3.7.2 : (i) Show that for any two timelike separated events $\Delta s^2 < 0$, there exists an intertial frame where the spatial separation $\Delta x = 0$, with $\Delta t \neq 0$. (ii) For two spacelike separated events $\Delta s^2 > 0$, show that there exists an inertial frame where $\Delta t = 0$ and $\Delta x \neq 0$.

3.7.3 Proper Time

For a timelike worldline we define the **proper time**, τ, by

$$\mathrm{d}\tau^2 = -\frac{\mathrm{d}s^2}{c^2}. \tag{3.7.9}$$

Thus

$$\mathrm{d}\tau = \frac{1}{c}\sqrt{-g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu} = \frac{\mathrm{d}\lambda}{c}\sqrt{-g_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\lambda}}. \tag{3.7.10}$$

In an inertial frame, if we choose co-ordinates so that $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2$, and choose $\mathrm{d}\lambda = \mathrm{d}t$, we get

$$\mathrm{d}\tau = \mathrm{d}t\sqrt{1 - \frac{1}{c^2}\left(\frac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t}\right)^2}. \tag{3.7.11}$$

If you are momentarily at rest with the particle, then $\mathrm{d}\boldsymbol{x} = 0$, and

$$\mathrm{d}\tau = \mathrm{d}t. \tag{3.7.12}$$

So proper time is simply the time measured by a clock moving along the worldline. It is a geometric, frame-independent quantity.

For a worldline connecting events P and Q, the proper time accumulated along the world line:

$$\tau_{PQ} = \int_P^Q \mathrm{d}\tau. \tag{3.7.13}$$

This is directly analogous to the length of a spatial curve that we discussed in the previous chapter. Another way to think about proper time is the following

It is the time measured on the hand watch of the observer moving on any trajectory (can be an accelerating observer). It measures the length of their worldline, hence it is a frame-independent, geometric quantity.

3.7.4 Free particle in spacetime

In a given inertial frame, what is the equation of motion of a particle moving along a timelike path ($\mathrm{d}s^2 < 0$ everywhere along the path)? For a free particle, we know that this should be a straight line in spacetime line $v(t - t_0) = (x - x_0)$ (ignoring the other two dimensions for simplicity). We can arrive at this known result by extremizing $\tau_{PQ} = \int_P^Q \mathrm{d}\tau$ for a collection of timelike paths passing through the events P and Q. To extremize

$$\tau_{PQ} = \int_P^Q \mathrm{d}\tau = \int_P^Q \sqrt{\mathrm{d}t^2 - \frac{\mathrm{d}x^2}{c^2}} = \int_P^Q \mathrm{d}\lambda \sqrt{\left(\frac{\mathrm{d}t}{d\lambda}\right)^2 - \frac{1}{c^2}\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2} \tag{3.7.14}$$

we are thinking of t and x as functions of some parameter λ, that is a parametrized trajectory $t(\lambda), x(\lambda)$. Then, extremizing the proper time for timelike trajectories would yield the following Euler Lagrange equations:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right) = 0, \quad \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{L}\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right) = 0, \quad \text{where} \quad L = \sqrt{\left(\frac{\mathrm{d}t}{\mathrm{d}\lambda}\right)^2 - \frac{1}{c^2}\left(\frac{\mathrm{d}x}{\mathrm{d}\lambda}\right)^2} \tag{3.7.15}$$

The EL equations will then yield $L^{-1}\mathrm{d}t/\mathrm{d}\lambda = const.$ and $L^{-1}\mathrm{d}x/\mathrm{d}\lambda = const.$ which implies $\mathrm{d}t/\mathrm{d}x = const.$ This is the equation of a straight line. So extremizing the proper-time gives us a way of getting the equation of motion of a free particle.

If I judiciously choose $\mathrm{d}\lambda = \mathrm{d}\tau$ (remember the straight line case in 2d plane), then we get $\mathrm{d}^2t/\mathrm{d}\tau^2 = \mathrm{d}^2x/\mathrm{d}\tau^2 = 0$. This is a nice simple form for the equation of a straight line in spacetime, it treats space and time symmetrically. We will often parametrize our curves in terms of the proper time along the curve. Note that the extremum is a maximum. Here is something you should keep in mind

For two timelike separated events P after Q, the straight line path provides the maximal proper time.

Why maximal? To understand this, think of an example two events at the same location in space in a given frame. These two can be connected by $c(t_P - t_Q) > 0$. Another trajectory made of two light-like segments moving away from Q and returning at P would yield a zero path length! Note that $\mathrm{d}\tau \leq \mathrm{d}t$. That is, the proper time interval is always shorter than (or equal to) the co-ordinate time interval.[4]

[4] Think about why this is not contradictory to what I said about proper time is maximized on straight line trajectories between two points.

Twin "Paradox" In the often discussed twin paradox (which as we see is not a paradox), it is common to write down the proper time of the traveling twin in terms of the proper time of the stationary one:

$$\tau(t) = \int_0^t d\bar{t}\sqrt{1 - \frac{1}{c^2}\left(\frac{dx}{d\bar{t}}\right)^2}. \tag{3.7.16}$$

Exercise 3.7.3 : Consider twins, one stationary and another with a wanderlust. Suppose the traveling twin travels on a straight line with speed $V = \alpha_1 c$. After a time $\tau_{\rm turn}$ as measured in the traveling twin's frame, the twin returns back to at speed $V = \alpha_2 c$. When the twins meet again, the traveling twin has aged by τ. Make a spacetime diagram showing their motion in the frame of the stationary twin. Find the age of the stationary twin at their reunion as a function of α_1, α_2, $t_{\rm turn}$ and τ. [This is a generalization of a Problem 4.9 in Hartle.] Set $\alpha_1 = \alpha_2 = \alpha$, and look at the result, does it make sense? You should note that the contribution to the age of the traveling twin from the infinitesimally small turnaround time is negligible in-spite of the large acceleration needed (see Example 4.3 of Hartle).

The expression for the time t in terms of the proper time of τ is given by

$$t(\tau) = \left[\left\{\int_0^\tau d\bar{\tau}\exp\left[\int_0^{\bar{\tau}} d\tilde{\tau}\frac{a(\tilde{\tau})}{c}\right]\right\}\left\{\int_0^\tau d\bar{\tau}\exp\left[-\int_0^{\bar{\tau}} d\tilde{\tau}\frac{a(\tilde{\tau})}{c}\right]\right\}\right]^{\frac{1}{2}}, \tag{3.7.17}$$

where $a(\tau)$ is the proper acceleration, which can be measured by the traveling twin in their own laboratory with an accelerometer [in the stationary twin's frame, it can be calculated as $\gamma^3 d^2x/dt^2$]. For a derivation of the above formula for $t(\tau)$, see here. Note that this allows the traveling twin to know what the time would be for the stationary one, completely based on their own local measurements (τ and $a(\tau)$).

Exercise 3.7.4 : (i) Use the previous exercise setup but now calculate t for the trip as a function of α and τ using the above expression (assume $\alpha_1 = \alpha_2 = \alpha$). Note that the acceleration is zero everywhere except at turnaround $\tau_{\rm turn}$ it can be modeled as a Dirac delta function in $a(\tau) = -c\ln[(1+\alpha)/(1-\alpha)]\delta_D(\tau - \tau_{\rm turn})$. Note that you could have determined the co-efficient of $\delta_D(\tau - \tau_{\rm turn})$ given the before and after trajectories with velocities $V = \pm\alpha c$, but I am giving it to you to make your life slightly easier. Hint: Make sure you break up the problem into before and after turnaround when evaluating each $\{\ldots\}$.[5] (ii) This is easier than (i), but do it anyways. Suppose that the traveling twin is on a trajectory with constant proper acceleration $a(\tau) = -g$ which returns them to their stationary twin after a proper time τ. Show that the time that passes for the stationary twin in terms of τ and g is $t = (2c/g)\sinh(g\tau/2c)$.

If you want to understand the twin paradox better, ie. why the traveling twin cannot claim to be stationary and hence reach the conclusion that the other twin must be younger when they

[5]Thanks to Miles for asking a relevant question that made me realize that the problem became harder than it was meant to be.

meet, first thing to realize that one of them is accelerating (which can be measured within their own frame), so they are not equivalent observers. This is true. However, one might argue that we can make the acceleration limited to a very small region of time (ie. think of piecewise two straightline trajectories, or even have a third person take over at the point of turnaround who has always been on the return trajectory). You will find that then the paradox resolves itself when you carefully think about the fact that the traveling twin's perspective does involve change from one inertial frame to another. Clarification of the "paradox" can be found by thinking of surfaces of simultaneity in the different frames. Ask questions like: From the point of view of the traveling twin, just before (one frame of traveling twin) and after (another frame of traveling twin) they change direction, what is the time in the stationary frame? The traveling twin will infer that the twin stationary is younger until the turnaround point. But after turnaround, the traveling twin infers that the stationary twin is older!

If the above word salad is confusing, here is a clean statement. The length of the wordlines of the stationary and traveling twin are both frame-independent, geometric quantities. Done.

3.8 Newtonian Physics as the Low-Speed Limit

For $V \ll c$,

$$d\tau = dt\sqrt{1-\frac{V^2}{c^2}} \simeq dt\left(1-\frac{V^2}{2c^2}+\cdots\right). \tag{3.8.1}$$

The relativistic free-particle action

$$S_{\text{free}} = -mc^2\int d\tau \tag{3.8.2}$$

therefore becomes

$$S_{\text{free}} \simeq -mc^2\int dt + \int dt\,\frac{mV^2}{2} + \cdots. \tag{3.8.3}$$

Apart from the irrelevant constant term, the familiar Newtonian kinetic action is recovered.

Thus Newtonian physics is not discarded. It is the low-speed limit of the spacetime geometry of Special Relativity.

3.9 Additional Practice Problems

1. Hartle Chapter 3, Problem 4
2. Hartle Chapter 3, Problem 5
3. Hartle Chapter 4, Problem 11
4. Hartle Chapter 4, Problem 14 from Hartle. Use Spacetime Diagrams.

3.10 Appendix

3.10.1 Einstein synchronization and the metric

Why does the metric take the particularly simple form

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2 \tag{3.10.1}$$

when we use the synchronized time of an inertial frame? Consider one spatial direction and allow, for the moment, the more general form

$$\mathrm{d}s^2 = -Ac^2\mathrm{d}t^2 + 2Bc\,\mathrm{d}t\,\mathrm{d}x + C\mathrm{d}x^2. \tag{3.10.2}$$

Einstein synchronization assigns time to clocks using the fact that light has the same one-way speed c in the $+x$ and $-x$ directions. Since light satisfies $\mathrm{d}s^2 = 0$, substituting $\mathrm{d}x = \pm c\mathrm{d}t$ gives

$$-A \pm 2B + C = 0. \tag{3.10.3}$$

The two signs imply $B = 0$ and $C = A$. We furthermore choose t to equal the proper time measured by clocks at rest in the reference frame. For such clocks $\mathrm{d}x = 0$, so $\mathrm{d}s^2 = -c^2\mathrm{d}t^2$, which fixes $A = 1$. Hence

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2. \tag{3.10.4}$$

In three spatial dimensions this becomes

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}S^2, \tag{3.10.5}$$

where $\mathrm{d}S^2$ is the Euclidean spatial metric, written in any time-independent spatial co-ordinates.

Other choices for synchronizing or labeling the clocks describe the same Minkowski spacetime, but the metric need not retain this simple form; in particular, space-time cross terms can appear.